

Ch 9: Light - Reflection and Refraction

Complete Notes | Deep Definitions | Formulae | Diagrams | Numericals | Centum Focus

SECTION A - BASIC IDEAS OF LIGHT AND SPHERICAL MIRRORS

1. Light

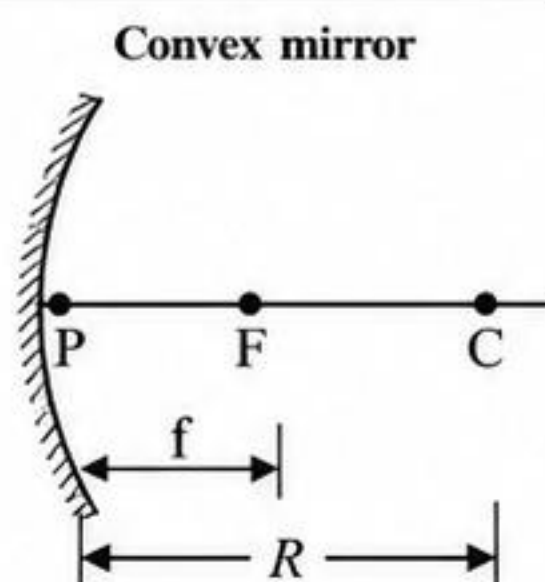
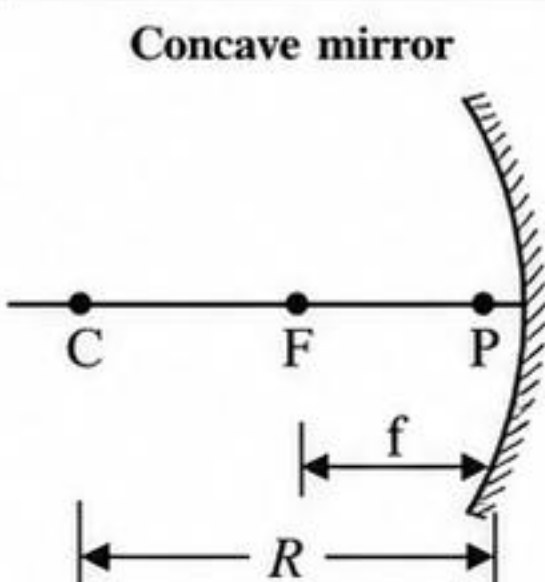
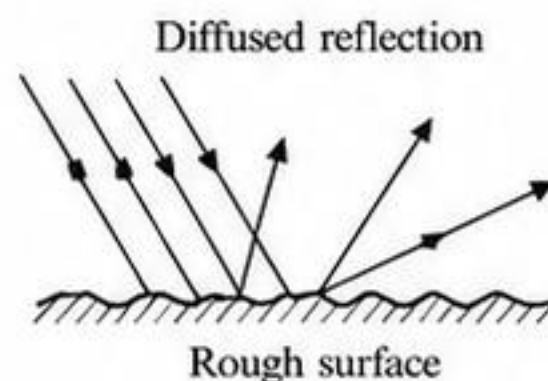
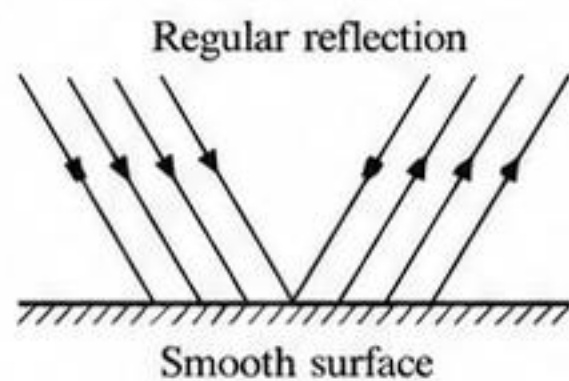
- Light is a form of energy that enables us to see objects.
- It travels in straight lines in a uniform medium.
- **Luminous objects:** Objects that produce their own light.
Examples: Sun, stars, candle, torch.
- **Non-luminous objects:** Objects that do not produce their own light but become visible when light falls on them.
Examples: Table, book, chair, moon.
- **Ray of light:** An ideal narrow path along which light travels in a straight line.
- **Beam of light:** A collection of a large number of light rays.
 - **Parallel beam:** Rays are parallel to each other.
 - **Converging beam:** Rays meet at a point.
 - **Diverging beam:** Rays spread out from a point.

2. Reflection of light

- Reflection of light is the bouncing back of light into the same medium when it strikes a surface.
- **Laws of reflection:**
 - (i) The angle of incidence (i) is equal to the angle of reflection (r).
 - (ii) The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.

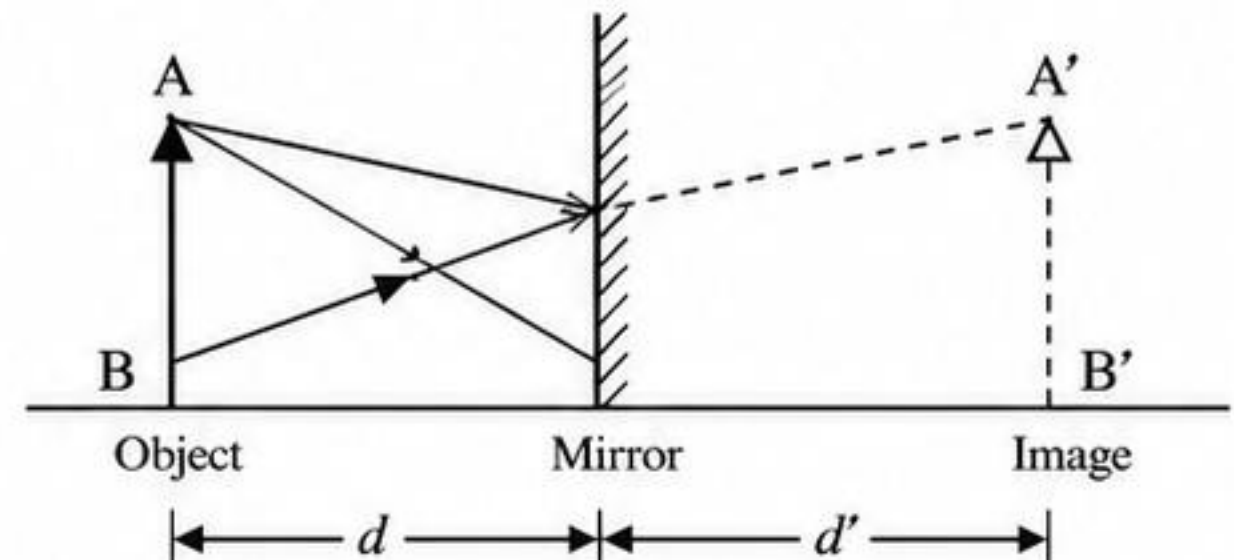
3. Types of reflection

- **Regular (specular) reflection:** It takes place on smooth, polished surfaces. The parallel incident rays remain parallel after reflection.
- **Diffused (irregular) reflection:** It takes place on rough surfaces. The parallel incident rays get reflected in different directions.



4. Image formed by a plane mirror

- A plane mirror always forms a **virtual** and **erect** image.
- The size of the image is the same as the size of the object.
- The image is formed at the same distance behind the mirror as the object is in front of the mirror.
- The image formed is laterally inverted (left appears right and right appears left).



d = distance of object from mirror
 d' = distance of image from mirror
For a plane mirror, $d = d'$

5. Spherical mirrors

- A **spherical mirror** is a mirror whose reflecting surface is a part of a sphere.
- **Types of spherical mirrors:**
 - **Concave mirror:** The reflecting surface is the inner (hollow) side of the spherical surface.
 - **Convex mirror:** The reflecting surface is the outer (bulging) side of the spherical surface.

6. Important terms related to spherical mirrors

- **Pole (P):** The midpoint of the reflecting surface of the mirror.
- **Centre of curvature (C):** The centre of the sphere of which the mirror is a part.
- **Radius of curvature (R):** The distance between the pole (P) and the centre of curvature (C). $R = PC$
- **Principal axis:** A straight line passing through the pole (P) and the centre of curvature (C).
- **Principal focus (F):** The point on the principal axis where the parallel rays, after reflection, either meet (concave) or appear to diverge from (convex).
- **Focal length (f):** The distance between the pole (P) and the principal focus (F). $f = PF$
- **Aperture:** The diameter of the reflecting surface.

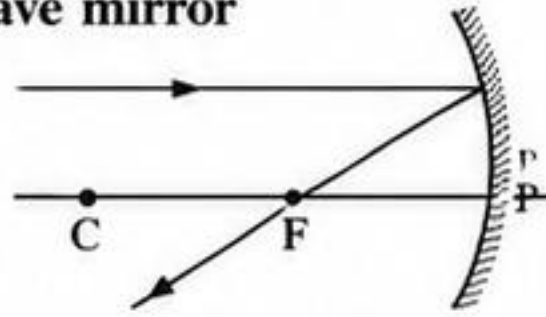
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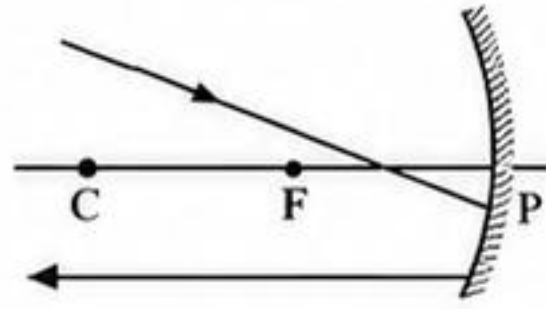
SECTION B - PRINCIPAL RAYS AND IMAGE FORMATION BY CONCAVE MIRROR

1. Principal rays for a concave mirror

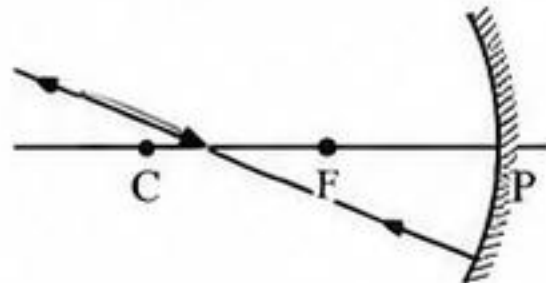
- A ray of light parallel to the principal axis, after reflection, passes through the focus (F).



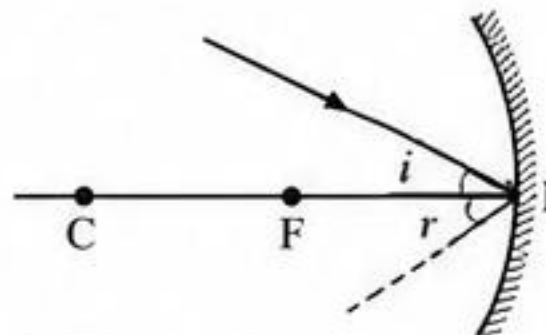
- A ray of light passing through the focus (F), after reflection, becomes parallel to the principal axis.



- A ray of light passing through the centre of curvature (C), after reflection, retraces its path.



- A ray of light striking the pole (P) of the mirror, after reflection, obeys the laws of reflection ($i = r$).



2. How ray diagrams are drawn

- Mark the principal axis and the pole (P) of the concave mirror.
- Mark the focus (F) and the centre of curvature (C) on the principal axis (with C farther from P: $PC = 2PF$).
- Draw the object (arrow) at the given position.
- From the tip of the object, draw at least two principal rays.
- After reflection, extend the rays (if needed). The point where the reflected rays meet (or appear to meet) gives the position of the image.
- Draw the image (arrow) at that point and describe its size and nature.

3. Image formation by a concave mirror

Position of object	Position of image	Size of image	Nature of image
At infinity	At F (focus)	Highly diminished (point-sized)	Real and inverted
Beyond C (centre of curvature)	Between C and F	Diminished	Real and inverted
At C (centre of curvature)	At C	Same size	Real and inverted
Between C and F	Beyond C	Enlarged	Real and inverted
At F (focus)	At infinity	Highly enlarged	Real and inverted
Between F and P (pole)	Behind the mirror	Enlarged	Virtual and erect

4. Representative ray diagrams

(i) Object beyond C (between infinity and C)

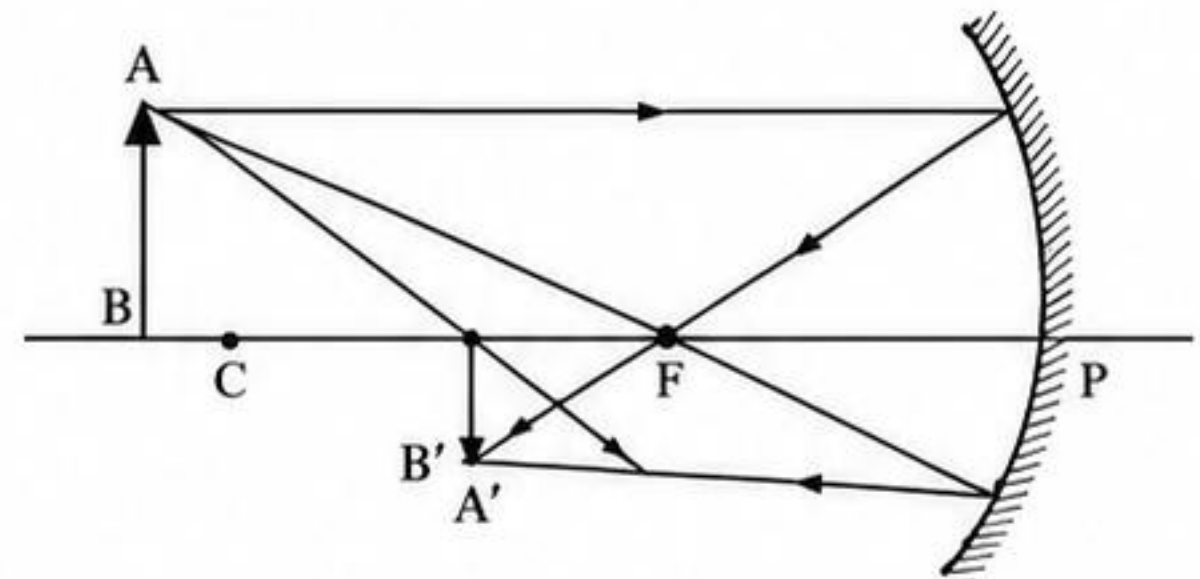


Image position: Between C and F

Size: Diminished

Nature: Real and inverted

(ii) Object between F and P

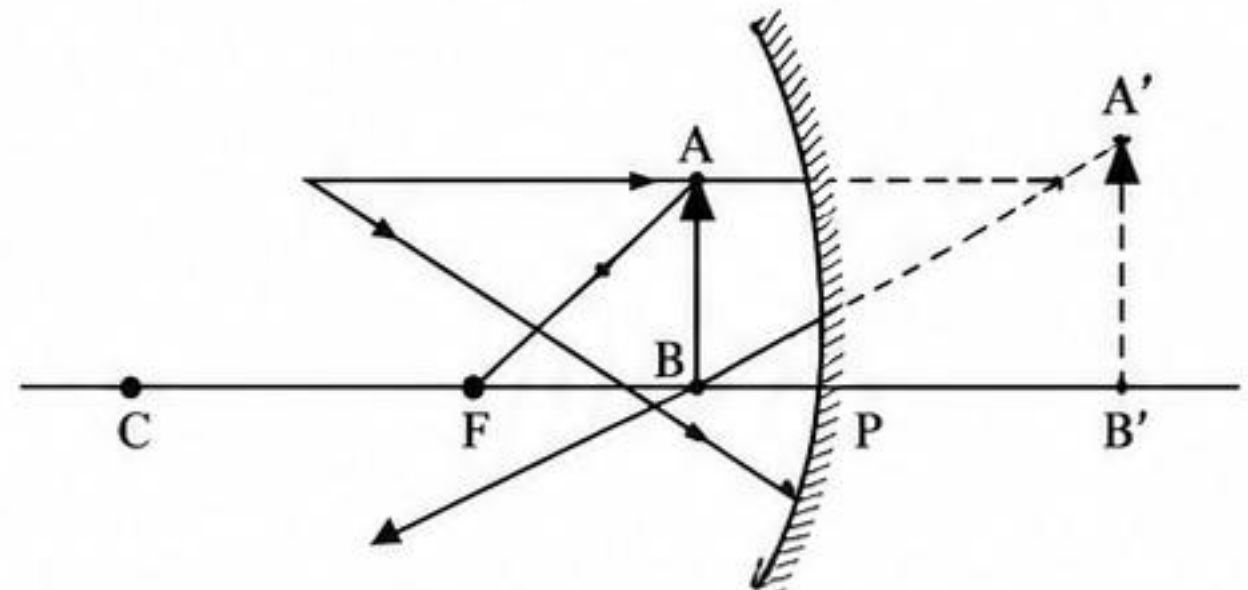


Image position: Behind the mirror

Size: Enlarged

Nature: Virtual and erect

5. Uses of concave mirror

- Shaving mirror:** Gives a magnified and erect image when the face is held close, helping in shaving.
- Dentist mirror:** Helps the dentist to see a magnified image of the teeth inside the mouth.
- Reflector in torches, headlights and searchlights:** Provides a parallel beam of light for long-distance visibility.
- Solar furnace:** Concentrates sunlight at the focus to produce very high temperatures for heating and experiments.

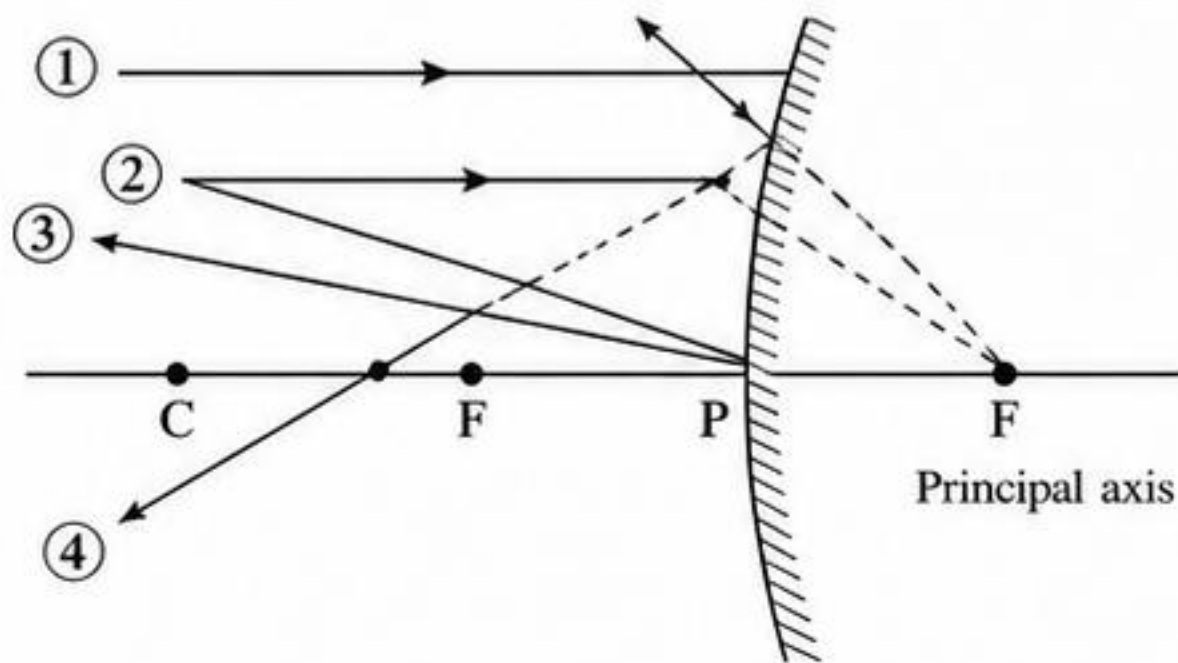
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SECTION C - CONVEX MIRROR, SIGN CONVENTION, MIRROR FORMULA AND MAGNIFICATION

1. Principal rays for a convex mirror

- **Ray parallel to the principal axis:** After reflection, it appears to diverge from the focus (F) behind the mirror.
- **Ray directed towards the focus (F):** After reflection, it becomes parallel to the principal axis.
- **Ray directed towards the centre of curvature (C):** After reflection, it retraces its path.
- **Ray incident at the pole (P):** After reflection, it obeys the laws of reflection (angle of incidence = angle of reflection).

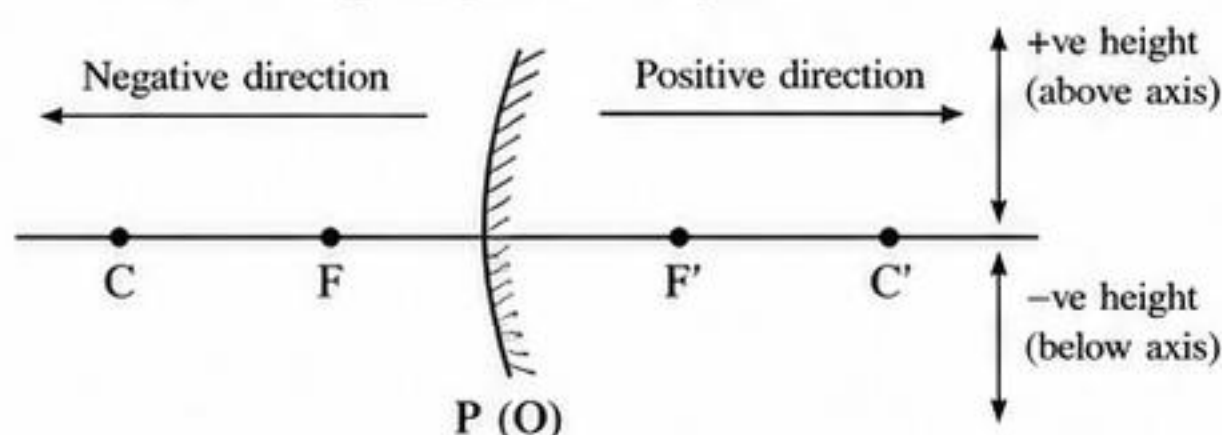


2. Image formation by a convex mirror

Position of object	Image position	Size of image	Nature of image
At infinity	At the focus (F) behind the mirror	Highly diminished (point-sized)	Virtual, erect
At any finite distance	Between the pole (P) and the focus (F) behind the mirror	Diminished	Virtual, erect

3. New Cartesian sign convention for spherical mirrors

- The pole (P) of the mirror is taken as the origin (O).
- All distances are measured along the principal axis.
- **Positive direction:** From the pole (P) along the principal axis towards the right.
- **Negative direction:** From the pole (P) along the principal axis towards the left.
- **Heights** measured perpendicular to the principal axis:
 - Above the principal axis: Positive.
 - Below the principal axis: Negative.



4. Mirror formula

- The mirror formula relates the object distance (u), image distance (v) and focal length (f):

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Where,

f = focal length of the mirror

v = image distance

u = object distance

- All distances should be used with their signs as per the new Cartesian sign convention.
- For a convex mirror: f is negative, v is negative, u is negative (for real objects in front).

5. Magnification for mirrors

- **Magnification (m)** is defined as:

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

Where,

h_i = height of the image

h_o = height of the object

- **Positive magnification:** Image is erect (upright).
- **Negative magnification:** Image is inverted.
- For a convex mirror, magnification is always positive and less than 1 (diminished image).

6. Worked example

A convex mirror has a focal length of 20 cm. An object is placed 40 cm in front of the mirror. Find the position and nature of the image.

Given: $f = -20$ cm (convex mirror), $u = -40$ cm

Using mirror formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{-20} = \frac{1}{v} + \frac{1}{-40}$$

$$\frac{1}{v} = -\frac{1}{20} + \frac{1}{40} = -\frac{1}{40}$$

$$v = -40 \text{ cm}$$

Magnification: $m = -\frac{v}{u} = -\frac{-40}{-40} = +1$

Since $m = \frac{h_i}{h_o}$ and $m = +1$, the image is erect and of same size as the object.

Answer: Image is formed 40 cm behind the mirror, it is virtual, erect and of the same size as the object.

7. Uses of convex mirror

- **Rear-view mirrors in vehicles:** Provide a wide field of view and erect, diminished images of vehicles behind.
- **Security mirrors:** Used in shops, corridors, and parking areas to view a large area and for safety and surveillance.

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SECTION D - REFRACTION OF LIGHT AND REFRACTIVE INDEX

1. Refraction of light

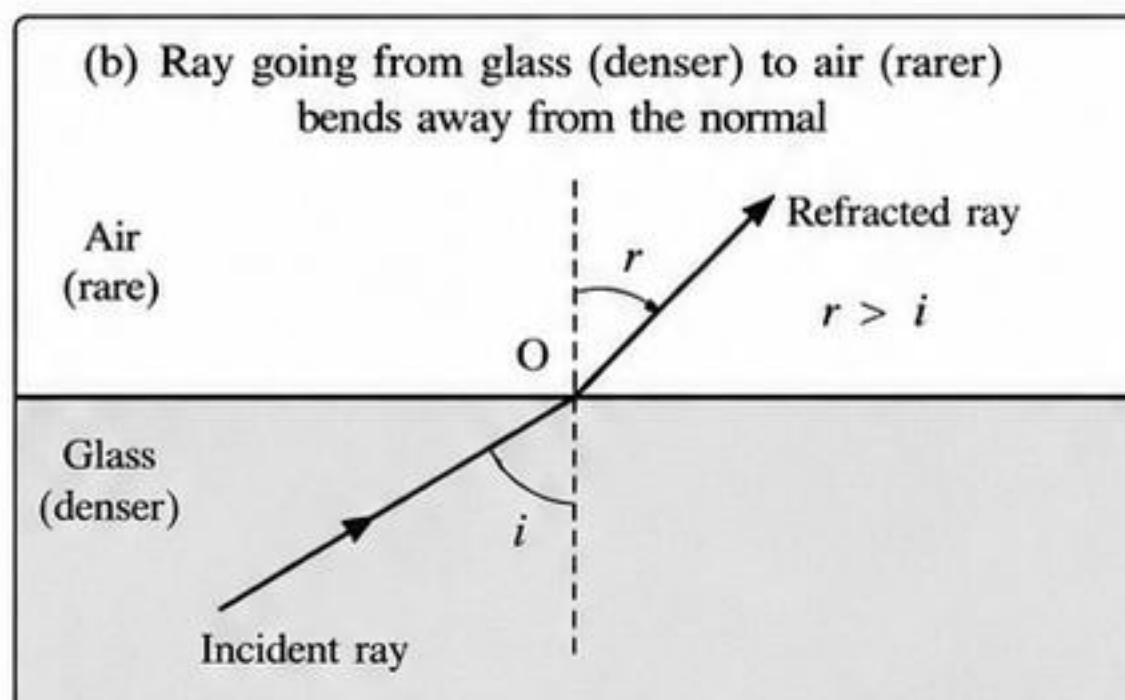
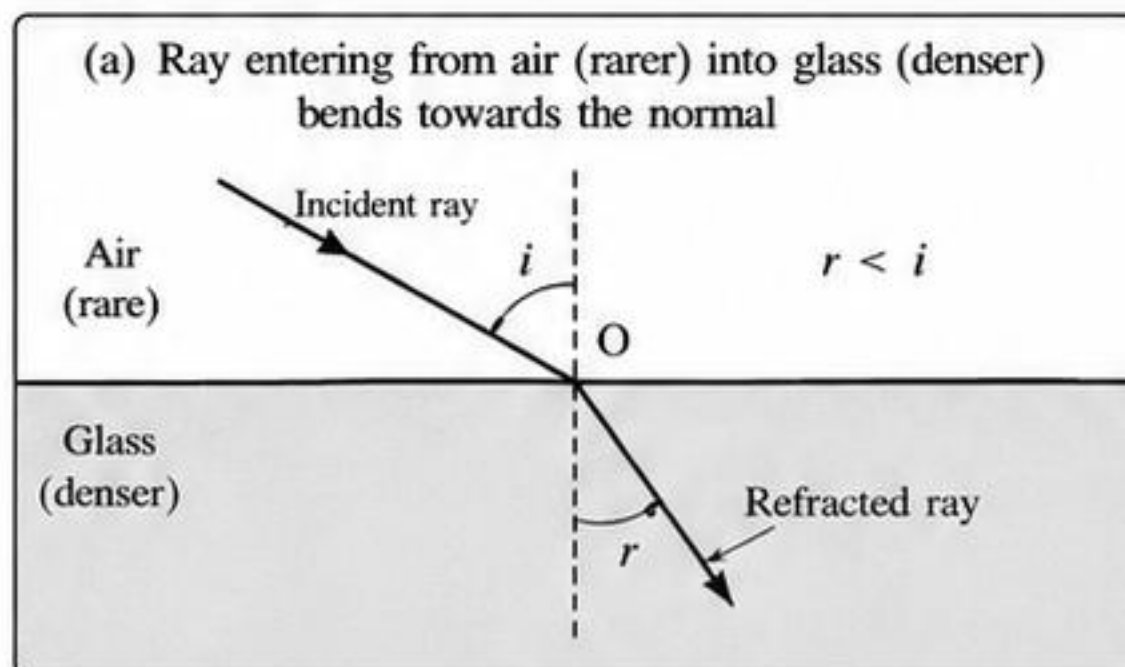
- Refraction is the change in direction of light when it passes obliquely from one transparent medium to another.
- It occurs because light travels with different speeds in different media. When light enters a new medium, the speed changes and the ray bends.

2. Optical density and speed of light

- **Optical density** of a medium is a measure of its ability to slow down light.
- A medium in which light travels slower is optically denser.
- **Relation:** The greater the optical density of a medium, the less is the speed of light in it.
- **Order:** Air (least dense) < Water < Glass < Diamond (most dense)

3. Refraction rules

- When light travels from a rarer (optically less dense) medium to a denser (optically more dense) medium, it bends towards the normal.
- When light travels from a denser medium to a rarer medium, it bends away from the normal.
- When light is incident normally ($i = 0^\circ$) on the surface separating two media, it does not bend; it continues in the same direction.



4. Laws of refraction (Snell's law)

- **First law:** The incident ray, refracted ray and the normal at the point of incidence all lie in the same plane.
- **Second law:** For a given pair of media, the ratio of the sine of the angle of incidence (i) to the sine of the angle of refraction (r) is constant.

$$n = \frac{\sin i}{\sin r} \quad (\text{for light going from air into a medium})$$

Where,

- i = angle of incidence
- r = angle of refraction
- n = refractive index (absolute) of the medium with respect to air

5. Refractive index (absolute)

- The absolute refractive index (n) of a medium with respect to air is the ratio of the speed of light in air (c) to the speed of light in that medium (v).

$$n = \frac{c}{v}$$

Where,

- c = speed of light in vacuum ($\approx 3.0 \times 10^8 \text{ m s}^{-1}$)
- v = speed of light in the medium
- n is always greater than 1 (for any material medium).

6. Relative refractive index between two media

- The refractive index of medium 2 with respect to medium 1 is the ratio of speed of light in medium 1 to that in medium 2.

$$n_{21} = \frac{v_1}{v_2}$$

Where,

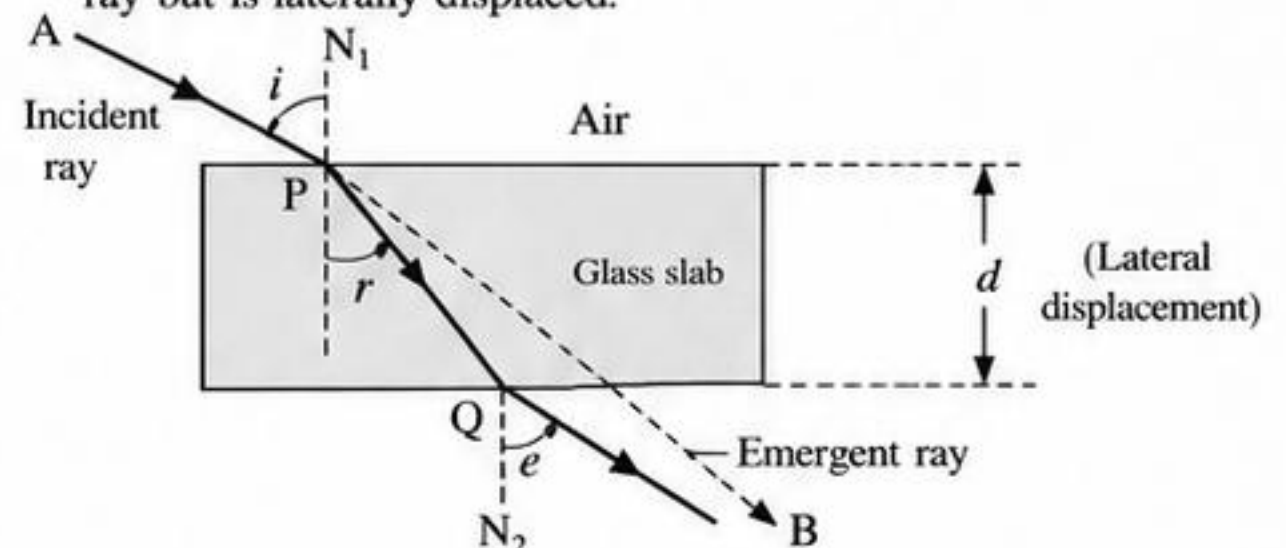
- v_1 = speed of light in medium 1
- v_2 = speed of light in medium 2

7. Real-life effects of refraction

- A pencil partly immersed in water appears to be bent at the water surface.
- The bottom of a pond or a swimming pool seems to be raised. This is called apparent depth.
- Objects under water appear nearer to the surface.

8. Lateral displacement through a glass slab

- When a ray of light passes through a rectangular glass slab with parallel faces, the emergent ray is parallel to the incident ray but is laterally displaced.



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SECTION E - SPHERICAL LENSES, TERMS AND PRINCIPAL RAYS

1. Definition of lens and spherical lens

- A **lens** is a transparent object bounded by two refracting surfaces, at least one of which is spherical.
- A **spherical lens** is a lens in which one or both of the refracting surfaces are spherical.

2. Types of lenses

- **Convex lens (converging lens):**
 - Thicker at the centre and thinner at the edges.
 - It converges a parallel beam of light to a point.
- **Concave lens (diverging lens):**
 - Thinner at the centre and thicker at the edges.
 - It diverges a parallel beam of light.

3. Important terms related to lenses

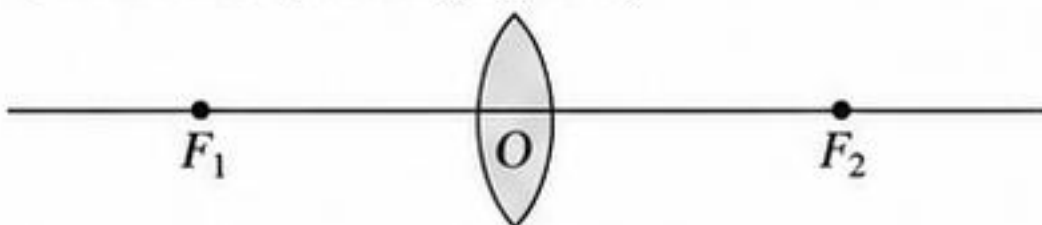
- **Optical centre (O):** The point at the centre of the lens through which a ray of light passes without any deviation.
- **Principal axis:** An imaginary straight line passing through the optical centre and the centres of curvature of the two surfaces of the lens.
- **Principal focus (F_1 and F_2):** The points on the principal axis on either side of the lens where parallel rays of light converge (convex lens) or appear to diverge from (concave lens).
- **Focal length (f):** The distance between the optical centre (O) and either principal focus. $f = OF_1 = OF_2$
- **Aperture:** The effective diameter of the lens.

4. Converging and diverging action

- **Convex lens (converging action):** A parallel beam of light incident on a convex lens converges to a point on the other side of the lens. Hence, it is called a converging lens.
- **Concave lens (diverging action):** A parallel beam of light incident on a concave lens diverges after refraction. The refracted rays appear to diverge from a point on the same side of the lens. Hence, it is called a diverging lens.

7. Diagrams of lenses

- **Convex lens (converging lens)**

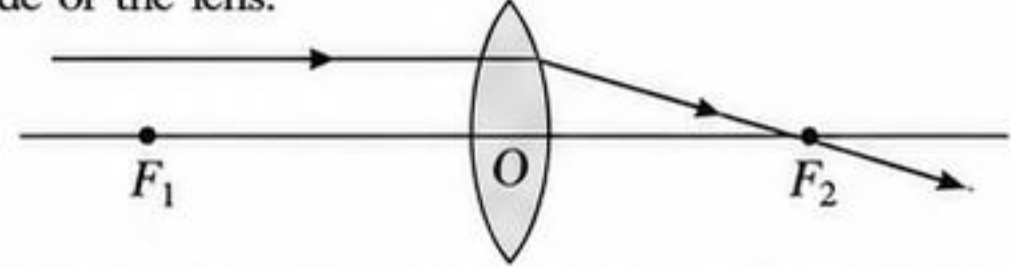


- **Concave lens (diverging lens)**

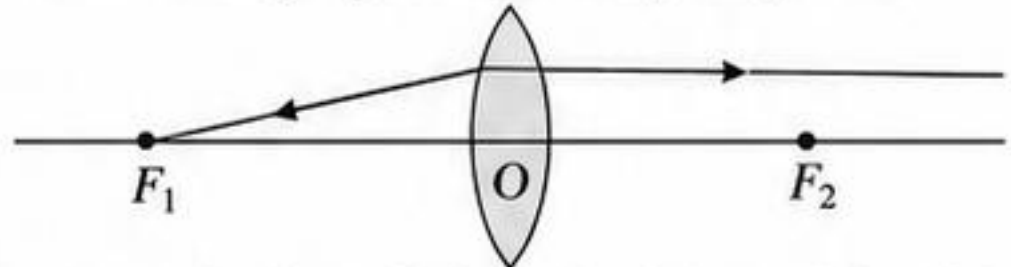


5. Principal rays for a convex lens

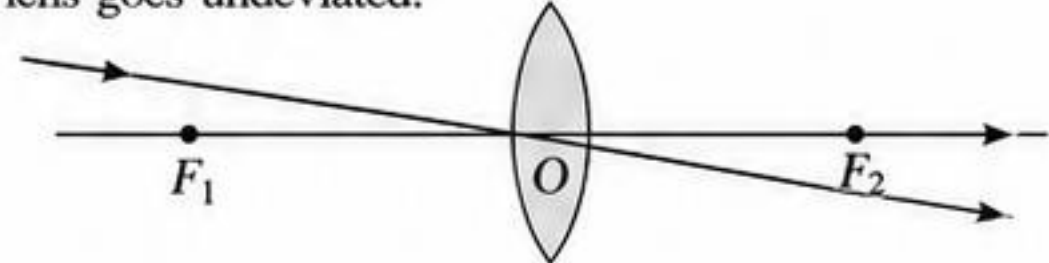
- A ray parallel to the principal axis after refraction passes through the principal focus F_2 on the other side of the lens.



- A ray passing through the principal focus F_1 after refraction emerges parallel to the principal axis.

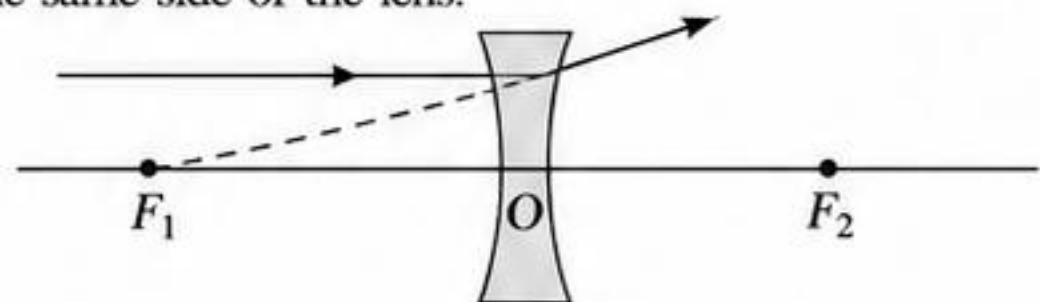


- A ray passing through the optical centre (O) of the lens goes undeviated.

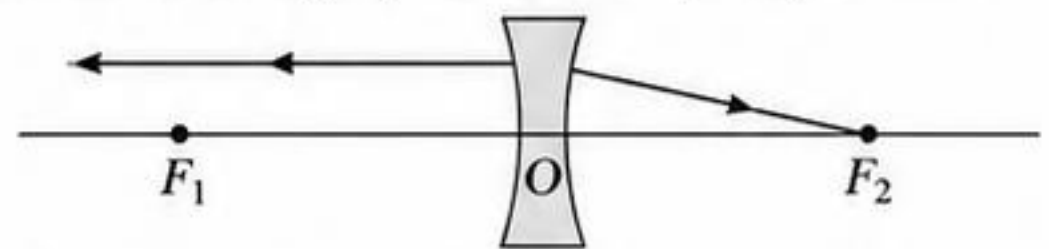


6. Principal rays for a concave lens

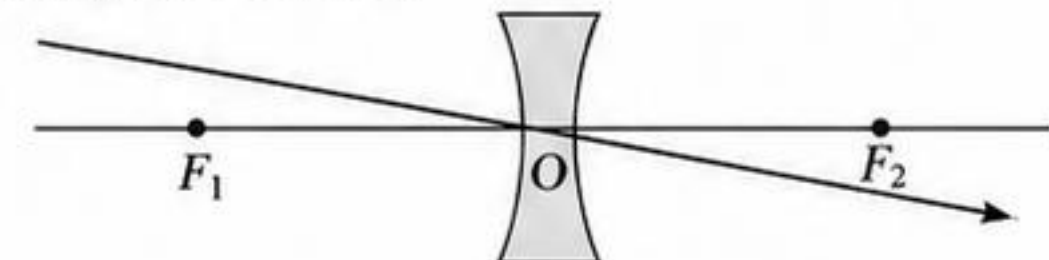
- A ray parallel to the principal axis after refraction appears to diverge from the principal focus F_1 on the same side of the lens.



- A ray directed towards the principal focus F_2 after refraction emerges parallel to the principal axis.



- A ray passing through the optical centre (O) of the lens goes undeviated.



8. Uses of lenses

- **Convex lens:** Used in magnifying glasses, microscopes, telescopes and camera lenses to form enlarged or real images of objects.
- **Concave lens:** Used in spectacles to correct myopia (short-sightedness) by diverging the incoming light and forming the image on the retina.

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SECTION F - IMAGE FORMATION BY LENSES, LENS FORMULA AND POWER

1. Image formation by a convex lens

Position of object	Image position	Size of image	Nature of image
At infinity	At F_2	Highly diminished	Real and inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real and inverted
At $2F_1$	At $2F_2$	Same size	Real and inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real and inverted
At F_1	At infinity	Highly enlarged	Real and inverted
Between O and F_1	On the same side of lens as the object	Enlarged	Virtual and erect

(F_1 is the object side focus, F_2 is the image side focus)

2. Image formation by a concave lens

Position of object	Image position	Size of image	Nature of image
At infinity	At F_1 (on the same side)	Highly diminished	Virtual and erect
At any finite distance	Between O and F_1 (on the same side)	Diminished	Virtual and erect

(F_1 is the focus on the object side of the concave lens)

3. Ray diagrams

(i) Convex lens : Object beyond $2F_1$

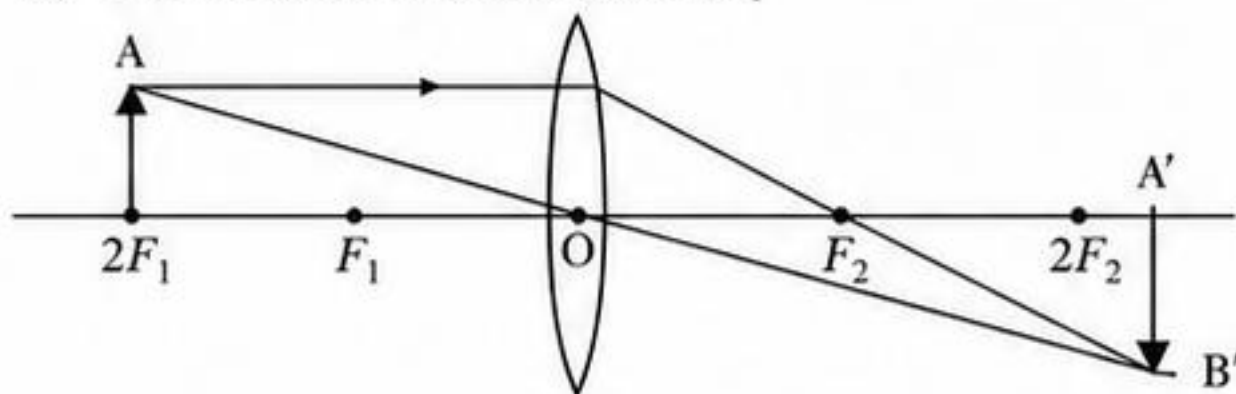


Image is formed between F_2 and $2F_2$, real, inverted and diminished.

(ii) Convex lens : Object between O and F_1

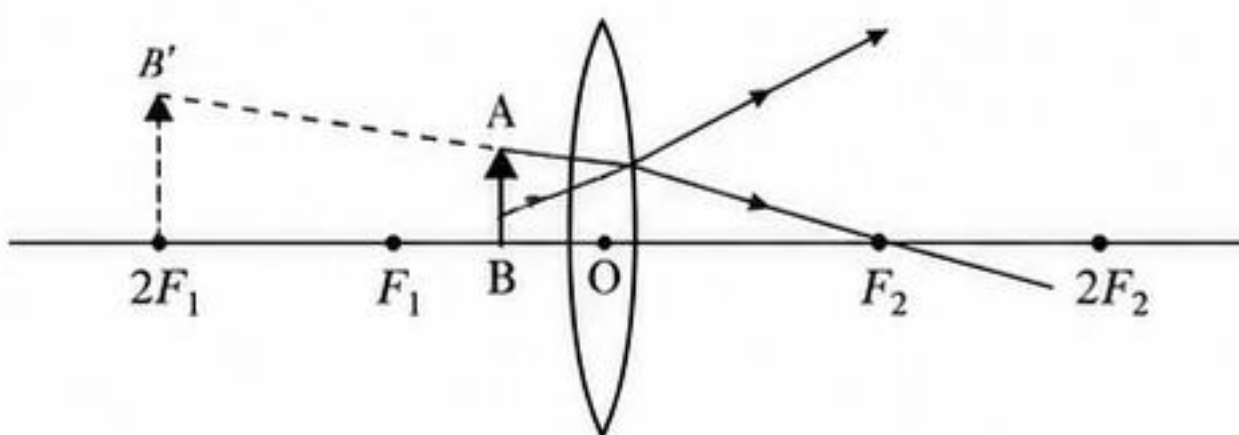


Image is formed on the same side of the lens as the object, virtual, erect and enlarged.

(iii) Concave lens : Object at any finite distance

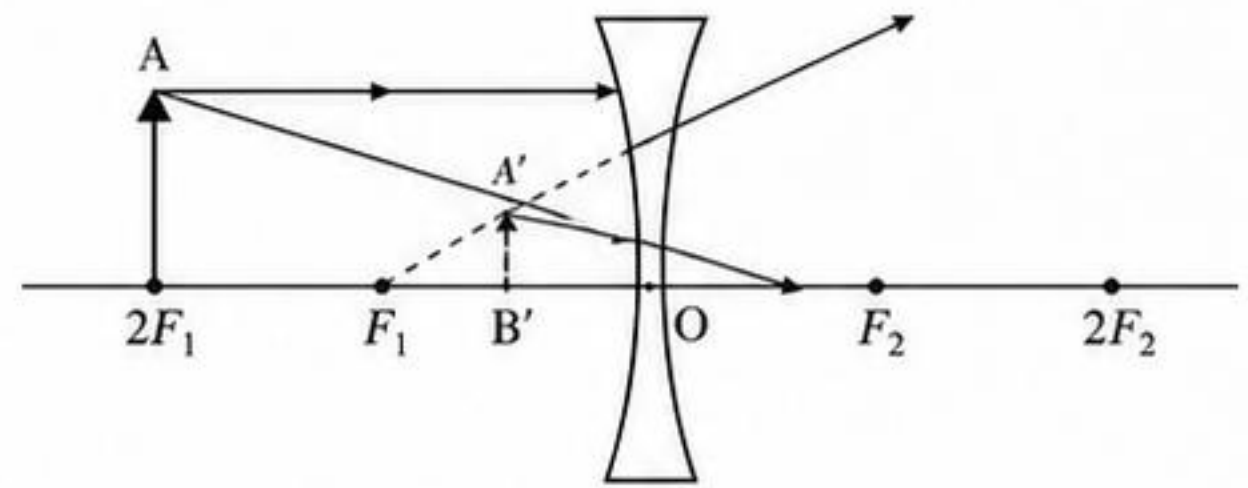


Image is formed between O and F_1 on the same side, virtual, erect and diminished.

4. Lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

where,

f = focal length of the lens (in metre)

v = image distance from the optical centre (in metre)

u = object distance from the optical centre (in metre)

Sign convention (New Cartesian convention):

- Distances measured in the direction of incident light (to the right) are taken as positive.
- Distances measured opposite to the direction of incident light (to the left) are taken as negative.

5. Magnification in a lens

$$m = \frac{h_i}{h_o} = \frac{v}{u}$$

where,

h_i = height of image, h_o = height of object

m is positive for erect image and negative for inverted image.

6. Power of a lens

$$P = \frac{1}{f} \quad (\text{with } f \text{ in metre})$$

- SI unit of power is dioptre (D).
- Power is positive for a convex lens and negative for a concave lens.

7. Worked example

Example: A convex lens has a focal length of 20 cm. Calculate (i) its power (ii) the image distance if the object is placed 60 cm in front of it.

Solution:

(i) $f = +20 \text{ cm} = +0.20 \text{ m}$

$$\text{Power } P = \frac{1}{f} = \frac{1}{0.20} = +5 \text{ D}$$

(ii) $u = -60 \text{ cm} = -0.60 \text{ m}$, $f = +0.20 \text{ m}$

$$\text{Using } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{0.20} = \frac{1}{v} - \left(\frac{1}{-0.60} \right) = \frac{1}{v} + \frac{1}{0.60}$$

$$\frac{1}{v} = 5 - 1.666... = 3.333... = \frac{10}{3} \Rightarrow v = \frac{3}{10} = +0.30 \text{ m} = +30 \text{ cm}$$

So, the image is formed 30 cm on the other side of the lens (real and inverted).

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SECTION G - IMPORTANT COMPARISONS, USES AND NUMERICAL TIPS

1. Comparison: Concave mirror vs Convex mirror

Feature	Concave mirror	Convex mirror
Reflecting surface	Inner (hollow)	Outer (bulging)
Nature of image	Can be real or virtual	Always virtual
Erect/Inverted	May be inverted or erect	Always erect
Size of image	May be magnified, diminished or same size	Always diminished
Field of view	Narrow	Wide
Focus (F)	In front of mirror	Behind mirror
Typical use	Shaving mirror, headlight, dentist mirror, solar cooker	Rear-view mirror, security mirror, traffic mirrors

2. Comparison: Convex lens vs Concave lens

Feature	Convex lens	Concave lens
Shape	Thicker at centre, thinner at edges	Thinner at centre, thicker at edges
Nature	Converging lens	Diverging lens
Effect on parallel rays	Converges at focus (real focus)	Diverges as if coming from focus (virtual focus)
Focal length (f)	Positive (+)	Negative (-)
Image formed	Can be real or virtual	Always virtual, erect and diminished
Typical use	Magnifying glass, camera, projector, spectacles (for hypermetropia)	Spectacles (for myopia), door viewer

3. Real image vs Virtual image

Basis	Real image	Virtual image
Formation	Formed by actual convergence of rays	Formed by apparent convergence (rays do not meet)
Screen	Can be obtained on a screen	Cannot be obtained on a screen
Nature	Always inverted	Always erect
Typical examples	Image by concave mirror (beyond C), convex lens (beyond F)	Image by plane mirror, convex mirror, concave mirror (between P and F), concave lens

7. Sign convention reminder (quick sketch)

For spherical mirrors

For lenses

- Distances measured along principal axis.
- Distances measured in direction of incident light are +ve; opposite direction are -ve.
- Heights above principal axis are +ve; below are -ve.
- Distances measured from optical centre (O).
- Distances to the right are +ve; to the left are -ve.
- Heights above principal axis are +ve; below are -ve.

4. Practical uses

- Concave mirror:**
 - Shaving/make-up mirrors (magnified face at close range)
 - Dentist's mirror (enlarged image)
 - Headlights/torches/solar cookers (parallel beam)
- Convex mirror:**
 - Rear-view mirrors in vehicles
 - Security mirrors in shops, ATMs, parking areas
 - Traffic mirrors at blind turns
- Convex lens:**
 - Magnifying glass (simple microscope)
 - Cameras, projectors, microscopes, telescopes
 - Spectacles for far-sightedness (hypermetropia)
- Concave lens:**
 - Spectacles for near-sightedness (myopia)
 - Door viewers (with convex lens combination)
 - To spread light in laser pointers and optical systems

5. Numerical problem-solving tips

- How to choose formula:**
 - Mirrors → Use mirror formula.
 - Lenses/Refraction → Use lens formula.
 - Size of image → Use magnification formula.
 - Medium comparison → Use refractive index formula.
 - Lens strength → Use power formula.
- How to apply sign convention:**
 - Always assign signs first (u, v, f, R, h, h_i).
 - For mirrors: follow mirror sign convention.
 - For lenses: follow Cartesian sign convention.
- How to predict image nature before calculation:**
 - Concave mirror:** Object beyond C → real, inverted, diminished.
Between C and F → real, inverted, magnified.
Between F and P → virtual, erect, magnified.
 - Convex mirror:** Always virtual, erect, diminished.
 - Convex lens:** Object beyond $2F$ → real, inverted, diminished.
Between $2F$ and F → real, inverted, magnified.
Within F → virtual, erect, magnified.
 - Concave lens:** Always virtual, erect, diminished.

6. Compact formula box

- Mirror formula:** $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
- Lens formula:** $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$
- Magnification:** $m = \frac{h_i}{h} = -\frac{v}{u}$
- Refractive index:** $\mu = \frac{\text{Speed of light in air/vacuum (c)}}{\text{Speed of light in medium (v)}}$
- Power of lens:** $P = \frac{1}{f(\text{in metres})}$ (unit: dioptre, D)

Ch 9: Light - Reflection and Refraction

Complete Notes | Deep Definitions | Formulae | Diagrams | Numericals | Centum Focus

SECTION H - HIGH-SCORE REVISION SHEET

1. Must-know definitions

- Reflection:** The bouncing back of light into the same medium when it strikes a surface.
- Refraction:** The change in direction of light when it passes from one transparent medium to another.
- Pole (P):** The midpoint of the reflecting surface of a spherical mirror.
- Principal focus (F):** The point on the principal axis where parallel rays, after reflection, meet (concave mirror) or appear to meet (convex mirror).
- Focal length (f):** The distance between the pole (P) and the principal focus (F).
- Centre of curvature (C):** The centre of the sphere of which the mirror is a part.
- Refractive index (n):** The ratio of the speed of light in vacuum to its speed in a given medium ($n = c/v$).
- Power of lens (P):** The reciprocal of the focal length (in metres), $P = 1/f$.
- Real image:** An image formed by actual convergence of light rays; can be obtained on a screen.
- Virtual image:** An image formed by apparent convergence of light rays; cannot be obtained on a screen.

2. Most important formulae (with symbols and SI units)

Formula (Name)	Expression	Symbols	SI Units
Mirror formula	$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	u : object distance (m) v : image distance (m) f : focal length (m)	metre (m)
Lens formula	$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$	u : object distance (m) v : image distance (m) f : focal length (m)	metre (m)
Magnification (Mirror)	$m = \frac{h_i}{h_o} = -\frac{v}{u}$	h_i : image height (m) h_o : object height (m) v : image distance (m) u : object distance (m)	dimensionless
Magnification (Lens)	$m = \frac{h_i}{h_o} = \frac{v}{u}$	h_i : image height (m) h_o : object height (m) v : image distance (m) u : object distance (m)	dimensionless
Refractive index (Snell's law form)	$n = \frac{\sin i}{\sin r}$	i : angle of incidence r : angle of refraction	dimensionless
Refractive index (Speed form)	$n = \frac{c}{v}$	c : speed of light in vacuum ($3.0 \times 10^8 \text{ m s}^{-1}$) v : speed of light in medium	dimensionless
Power of lens	$P = \frac{1}{f}$	P : power (diopetre, D) f : focal length (m)	diopetre (D)

3. Most important differences

Aspect	Regular (Specular) Reflection vs Diffused (Irregular) Reflection		Concave Mirror vs Convex Mirror	Convex Lens vs Concave Lens	Reflection vs Refraction	
Surface / Medium	Smooth, polished surface vs Rough, uneven surface		Inner (hollow) surface vs Outer (bulging) surface	Thicker at centre vs Thinner at centre	Same medium vs Different medium	
Behaviour of rays	Parallel rays remain parallel	Parallel rays get scattered in many directions	Converging mirror vs Diverging mirror	Converging lens vs Diverging lens	Bouncing back of light vs Change in direction of light	
Focus nature	Not applicable		Real focus vs Virtual focus	—	—	
Image formed	Usually clear images	No clear image	Can form real & virtual images (size & nature depend on position)	Can form real & virtual images (size & nature vary)	Image may form	No image forms (only path changes)
Example	Mirror, calm water	Paper, wall, unpolished metal	Spoon (inner) vs Back of spoon (outer)	Magnifying glass vs Spectacles (diverging)	Reflection in a mirror vs Bending of light in water	

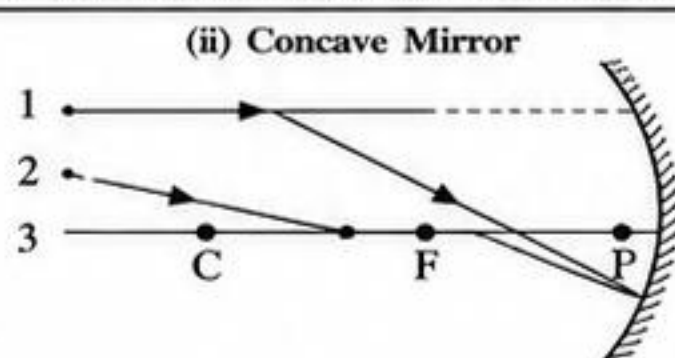
4. Common facts and ray rules

- Laws of reflection:** (i) $i = r$, (ii) incident ray, reflected ray and normal at the point of incidence lie in the same plane.
- Rays parallel to principal axis:** (i) Concave mirror $\rightarrow$ pass through F. (ii) Convex mirror $\rightarrow$ appear to come from F (behind mirror).
- Rays through F:** (i) Concave $\rightarrow$ emerge parallel to principal axis. (ii) Convex $\rightarrow$ emerge diverging as if from F.
- Rays through C:** After reflection, retrace their path.
- Rays through P:** Reflect symmetrically (angle with axis is reversed).
- For lenses:** (i) Parallel rays $\rightarrow$ through F (convex), appear from F (concave). (ii) Through F $\rightarrow$ emerge parallel. (iii) Through optical centre $\rightarrow$ pass undeviated.
- Real images are inverted; virtual images are erect.
- In denser medium, speed decreases and light bends towards normal.

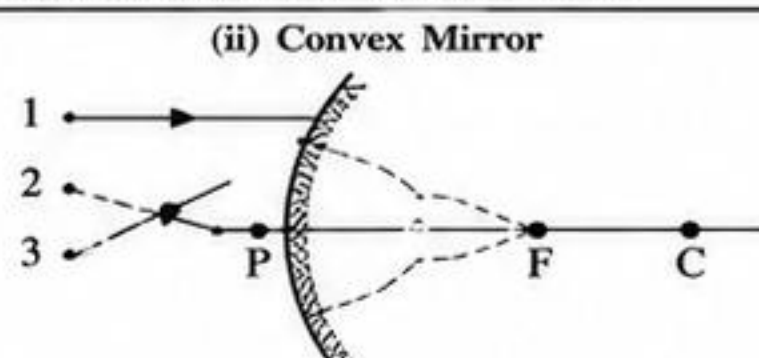
5. Centum final points (Exam tips)

- Learn all definitions word-for-word.
- Write formulae neatly with proper symbols and units.
- Remember sign convention: u is negative for real object, positive for virtual object; v is positive for real image (lens) or negative for real image (mirror); f is negative for concave mirror/concave lens.
- Draw neat ray diagrams with proper labels (P, F, C, O).
- Mention nature, position and size of image wherever asked.
- Always show construction rays clearly in numericals.
- For lens power, keep f in metres and answer in dioptres (D).
- State laws/ray rules before using them in the answer.
- In refraction problems, decide denser and rarer medium first.
- Practice typical numericals: mirror formula, lens formula, magnification, power of lens and refractive index ($n = \sin i / \sin r$, $n = c/v$).

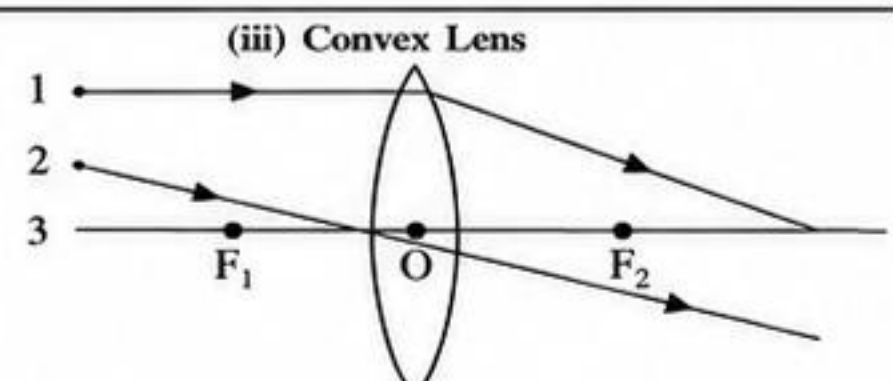
6. Mini-summary diagram strip: Principal rays (quick visual recall)



1. Parallel to axis $\rightarrow$ through F
2. Through F $\rightarrow$ parallel to axis
3. Through C $\rightarrow$ retraces its path



1. Parallel to axis $\rightarrow$ appears to come from F
2. Directed towards F $\rightarrow$ reflects parallel to axis
3. Directed towards C $\rightarrow$ retraces its path



1. Parallel to axis $\rightarrow$ through F_2
2. Through F_1 $\rightarrow$ parallel to axis
3. Through O $\rightarrow$ undeviated